



GREEN UNIVERSITY OF BANGLADESH

Industrial Training

on

User Interface Design and Frontend Implementation of a Web-Based Point of Sale (POS) Management System for O'Dell Tech Ltd.

Submitted by

Anisur Rahaman Maruf

222902078

*An industrial training report submitted to the Department of Computer Science & Engineering,
for the partial fulfillment of the degree of Bachelor of Science in Computer Science & Engineering*

Under the Supervision of

Ms. Mayeesha Farjana

Lecturer, Department of Computer Science and Engineering

Training Place

O'Dell Tech Ltd.

Road 10, Avenue 11, Mirpur DOHS, Dhaka

Training Duration : 16 August 2026 to 19 September 2026

Industrial Supervisor : Nusrat Jahan, Data Scientist

Report Submission Date : 28 September 2026

Summer 2026

Declaration

I declare that this industrial training report reflects my original work, observations, and technical learning completed during my internship at O'Dell Tech Ltd. . It specifically highlights my contributions to the user interface design, layout structure, and frontend development of the web-based **Point of Sale (POS) system** . Any material, ideas, or references drawn from external research or standard documentation have been duly acknowledged .

Furthermore, I confirm that this report has not been submitted, either in whole or in part, to any other university or institution for the award of any degree or academic qualification . I accept full responsibility for the accuracy and originality of the account described as my own .

Anisur Rahaman Maruf

ID: 222902078

Dept. of CSE

Green University of Bangladesh

Certificate

This is to certify that the industrial training report entitled *User Interface Design and Frontend Implementation of a Web-Based **Point of Sale (POS)** Management System for O'Dell Tech Ltd.* has been prepared and submitted by **Anisur Rahaman Maruf** in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering on 19 September 2026.

Ms. Mayeesha Farjana
Academic Supervisor
Lecturer, Dept. of CSE
Green University of Bangladesh

Nusrat Jahan
Industrial Supervisor
Data Scientist
O'Dell Tech Ltd.

*Accepted and approved in partial fulfillment of the requirements for the degree
Bachelor of Science in Computer Science and Engineering.*

Mr. Syed Ahsanul Kabir
Chairperson, Dept. of CSE
Green University of Bangladesh

Ms. Rokeya Khatun
Lecturer, Dept. of CSE
Green University of Bangladesh

Ms. Sagufta Sabah Nakshi
Lecturer, Dept. of CSE
Green University of Bangladesh

Acknowledgments

Acknowledgement

First and foremost, I express my gratitude to Almighty Allah for giving me the strength and opportunity to complete this industrial training.

I would like to express my sincere gratitude to all those who supported and guided me throughout the industrial training period from **16 August 2026** until **19 September 2026**. Their encouragement and assistance were invaluable, especially during times when I faced challenges.

I am thankful to my academic mentor, **Ms. Mayeesha Farjana, Lecturer, Department of Computer Science and Engineering, Green University of Bangladesh**, for providing consistent guidance and support during the training. Their feedback and suggestions helped me stay focused and improved the overall quality of my learning experience.

I am also grateful to the professionals at **O'Dell Tech Ltd.**, especially my industrial supervisor **Nusrat Jahan, Data Scientist**, for their valuable insights, technical guidance, and for sharing their industry experience. Their support greatly enhanced my understanding of real-world practices, tools, and work culture.

Finally, I would like to thank **my family, friends, and classmates** for their constant encouragement, emotional support, and motivation, which played a vital role in helping me successfully complete the training.

Abstract

This report summarizes my industrial internship experience at **O'Dell Tech Ltd.** from **16 August 2026** to **19 September 2026**, completed as part of the requirements for the degree of **Bachelor of Science in Computer Science and Engineering** at **Green University of Bangladesh**. My primary assignment was contributing to the design and development of a web-based **Point of Sale (POS) Management System** for retail operations.

For the POS project, my primary role focused on **website design, user interface (UI/UX) layout, and frontend implementation**. The core technologies used included **HTML5, CSS3, JavaScript, Bootstrap, and Git/GitHub** for version control, seamlessly integrated within Django's **Model-View-Template (MVT)** architecture.

My technical contributions centered on crafting an intuitive, responsive, and user-friendly experience across all core POS modules. Key activities included designing the system **Dashboard**, the interactive **New Sale** interface, the dynamic **Shopping Cart**, **Customer and Employee Management** views, and printable **Invoice layouts**. I worked on implementing product search, real-time client-side price and discount display, and maintaining a clean visual layout across the system.

Overall, this internship bridged the gap between academic theory and real-world software engineering, significantly strengthening my competencies in frontend development, UI design, team collaboration, and technical problem-solving.

Keywords: Industrial Internship, Point of Sale (POS), Frontend Development, User Interface (UI/UX) Design, HTML/CSS/JavaScript, Bootstrap.

TABLE OF CONTENTS

Declaration	i
Certificate	ii
Acknowledgments	iii
Abstract	iv
List of Figures	x
List of Tables	xi
1 Introduction	1
1.1 Problem Domain	1
1.2 Motivation of Training	2
1.3 Objective of Training	2
1.4 Scope of Training	3
1.5 Identifying the Gap Between Academia and Industry	4
1.6 Challenges and Areas for Growth	5
1.6.1 Challenges	5
1.6.2 Areas for Growth	5
1.7 Industrial Training Details Plan	6
1.7.1 Training Resource Plan	6
1.7.2 Training Time Management Plan	6
1.8 Risk Analysis and Resolve Process	7
1.9 Industrial Training Report Outline	7
1.10 Conclusion	8

2	Organization Overview	9
2.1	Introduction	9
2.2	Company Profile	9
2.2.1	Historical Timeline	10
2.2.2	Work Environment.....	11
2.2.3	Mission and Vision.....	12
2.2.4	Core Values	12
2.3	Organizational Structure.....	13
2.3.1	Roles and Responsibilities of Officials.....	13
2.3.2	Collaboration and Interdepartmental Coordination	15
2.4	Products/Services	15
2.4.1	Human Resources Management	15
2.4.2	Merchandising and Production Management	15
2.4.3	Production Sample Management	15
2.4.4	Web Design and Development	16
2.4.5	Artificial Intelligence and Machine Learning	16
2.4.6	Server Management	16
2.4.7	Software, Applications and Quality Assurance	16
2.4.8	Innovation Hub and Training	16
2.5	Clients	17
2.6	Direct Involvement of Resources During Training.....	17
2.7	Constructive Feedback About the Training Program	18
2.8	Conclusion.....	19
3	Specific Details on Training Activities and Report	20
3.1	Introduction	20
3.2	Weekly Duties and Activities	21
3.2.1	First Week: Orientation, Requirement Review, and UI Planning (16–20 August 2026)	21

3.2.2	Second Week: Django Templating and Practice Application (23–27 August 2026).....	21
3.2.3	Third Week: POS Interface Design and Module Layouts (30 August – 2 September 2026)	22
3.2.4	Fourth Week: Advanced UI Engineering, Cart Logic, and Test- ing (8–13 September 2026).....	23
3.2.5	Fifth Week: Deployment Verification, QA, and Final Report (14–19 September 2026)	24
3.3	Key Projects	24
3.3.1	POS Project Scope and My Specific Contributions	24
3.3.2	Technologies and Development Tools	25
3.3.3	System Organization and UI Development Sequence	25
3.3.4	User Interface Implementation Details	26
3.3.5	Client-Side Testing and UI Validation.....	34
3.3.6	Other Responsibilities.....	34
3.3.7	Key Performance Indicators (KPIs).....	35
3.3.8	Internship Summary	36
4	Acquired Skills During Industrial Training.....	38
4.1	Introduction	38
4.2	Learning New Technologies and Applying Them to Solve Engineering Problems.....	38
4.3	Theoretical Knowledge Gained.....	40
4.3.1	UI/UX Design Principles & Human-Computer Interaction (HCI)	40
4.3.2	Software Testing Principles for User Interfaces.....	41
4.3.3	Frontend-Backend Coupling in Modern Web Architecture	41
4.4	Practical Knowledge Acquired.....	41
4.4.1	Hands-on UI/UX Development Experience.....	41
4.4.2	Defect Tracking and Client-Side Debugging.....	41
4.4.3	Practical Perspective on Starting a Tech Startup.....	42

4.5	Ethical Responsibilities and Professional Conduct	42
4.6	Conclusion.....	43
5	Conclusion.....	44
5.1	Summary	44
5.2	Challenges During Industrial Training	45
5.2.1	Understanding Requirements and UI/UX Integration.....	45
5.2.2	Client-Side Interactivity and Testing	45
5.2.3	Time Management and Cross-Team Communication.....	45
5.2.4	Deployment and UI Verification	45
5.3	Scope of the Study.....	46
5.4	Problems and Executions	46
5.5	Overall Achievement.....	47
	Annexure: PO-Skill Mapping.....	48
	References	57

List of Figures

1.1	Time Management of the Industrial Training Process	6
1.2	Four Steps of the Risk Management Process.....	7
2.1	Historical Timeline of O’Dell Tech.....	11
2.2	O’Dell Tech Organizational Chart	13
3.1	SDLC Stages	21
3.2	Main functional groups of the POS system	23
3.3	Detailed POS development sequence	25
3.4	POS System Login Interface	26
3.5	POS System Dashboard.....	26
3.6	Employee Management Interface	27
3.7	New Sale Product Catalog and Search Interface	28
3.8	Interactive Shopping Cart and Checkout Overview.....	28
3.9	Customer Management Interface.....	29
3.10	Printable Sales Invoice Interface	29
3.11	Product Management Interface.....	30
3.12	Bulk Product Addition Interface.....	30
3.13	Category Management Interface.....	31
3.14	Supplier Management Interface.....	31
3.15	Inventory Management Interface	32
3.16	Report Overview	32
3.17	Sale Report Overview.....	33
3.18	User Activity Log	33

3.19	POS System GitHub Deployment	34
------	------------------------------------	----

List of Tables

3.1	Django Templating and Practice Activities	22
3.2	Recorded Development and Testing Milestones.....	35
3.3	Weekly Recorded Working Hours	36
3.4	Internship Summary	37
4.1	IEEE Code of Ethics Applied During Industrial Training	43
5.1	Program Outcome (PO) Fulfillment in Industrial Training	49
5.2	Complex Engineering Problems (WP) – Application to Project.....	50
5.3	Complex Engineering Activities (EA) – Application to Project.....	52
5.4	Mapping of All 17 SDG Goals with Achievements from Industrial Training	54

Chapter 1

Introduction

Industrial training is a fundamental component of the **Bachelor of Science in Computer Science and Engineering (CSE)** curriculum at **Green University of Bangladesh**. It bridges classroom academic knowledge with real-world professional software engineering practices. As part of this mandatory requirement, I completed my industrial internship at **O'Dell Tech Ltd.** from **16 August 2026** to **19 September 2026**.

My primary assignment during this period was contributing as a **Frontend Developer** and **UI/UX Designer** to the design and implementation of a web-based **Point of Sale (POS) Management System** for retail enterprise operations. This chapter presents the background, domain analysis, motivation, objectives, scope, risk management strategies, and operational timeline of my industrial training.

1.1 Problem Domain

In modern retail management, point-of-sale systems require fast, error-free, and highly intuitive user interfaces to ensure smooth counter operations, minimize customer checkout waiting times, and reduce human error. Traditional POS systems often suffer from cluttered screens, non-responsive layouts, and complex navigation structures that hinder cashier efficiency.

Developing a web-based POS application required bringing product catalogs, customer data, sales processing, cart adjustments, invoice generation, and real-time inventory displays into a unified, responsive frontend interface. My specific focus within this domain was engineering a

seamless **user interface (UI)** and **user experience (UX)** using **HTML5, CSS3, JavaScript**, and **Bootstrap 5**, tightly integrated with Django's **Model-View-Template (MVT)** web architecture. The challenge was to ensure that dynamic client-side interactions such as **live product filtering, instant price and discount calculations, responsive shopping cart updates, and printable invoice styling** performed efficiently across diverse desktop and tablet screen resolutions.

1.2 Motivation of Training

The primary motivation behind undertaking this industrial training was to transition from theoretical academic projects to building a production-ready, business-critical user interface within an established software company. Specific motivations included:

- **Real-World UI/UX Application:** Applying theoretical human-computer interaction and web design principles to build a commercial POS interface used in fast-paced retail operations.
- **Frontend-Backend Integration:** Understanding how modern client-side technologies (**HTML/CSS/JS/Bootstrap**) dynamically communicate with Django views, context dictionaries, and database ORM layers.
- **Industry Workflows & Standards:** Experiencing professional team collaboration, **Git/GitHub** version control workflows, agile task distribution, and client-side debugging under strict delivery schedules.
- **Bridging User Needs and Technical Logic:** Translating complex business rules such as tiered customer discounts, tax rates, and stock alerts into clear, visual UI feedback for end-users.

1.3 Objective of Training

The core objectives of my industrial training at **O'Dell Tech Ltd.** were as follows:

- **Frontend Engineering:** Design, build, and optimize responsive, user-friendly templates for core POS modules including the **Dashboard, New Sale interface, Interactive Cart, Customer Management, Employee Management, and Sales Invoices.**
- **Client-Side Interactivity:** Implement client-side **JavaScript** logic for instant product search, real-time discount/tax calculations, and cart adjustments without requiring full page reloads.
- **Responsive & Consistent Styling:** Utilize **Bootstrap grid systems** and custom **CSS** to ensure layout consistency, clean typography, and accessibility across various user roles (**Cashier, Manager, Admin**).
- **Integration & Version Control:** Collaborate with backend team members through **Git and GitHub** to integrate frontend templates with Django views and database structures.
- **Quality Assurance & Testing:** Perform client-side debugging, **UI/UX evaluation**, cross-browser compatibility testing, and retesting following defect resolution.

1.4 Scope of Training

The scope of this industrial training encompassed structured learning tasks, foundational web assessments, and active frontend development on the primary POS application:

In-Scope Activities:

- Orientation and review of **Business Requirement Specifications (BRS).**
- Preliminary skill assessments in **Python, Django architecture, and frontend fundamentals**, including building a practice task-management application.
- Designing **interface wireframes** and **Bootstrap layouts** for all primary POS screens.
- Frontend implementation of **live product searching, shopping cart modifications, customer identification displays, and custom invoice layouts.**
- Styling product import interfaces, data tables, filter forms, and navigation sidebars.

- **Client-side bug fixing, UI polish, cross-browser retesting,** and deployment verification on **pos.odelltech.com**.

Out-of-Scope Activities:

- Backend database design, server-side ORM query optimization, hardware deployment, and non-software business operations of **O'Dell Tech Ltd.**

1.5 Identifying the Gap Between Academia and Industry

Academic coursework at **Green University of Bangladesh** provided a strong foundation in computer science fundamentals, basic programming, web development principles, and software engineering methodologies. However, working on a commercial POS project highlighted significant differences between academic projects and industrial software development:

- **Static Assignments vs. Dynamic Application:** University assignments typically focus on static web pages or isolated backend scripts. In industry, the frontend must dynamically render complex, interconnected business data (e.g., live stock status, historical discounts, tax calculations) in real time.
- **UI/UX Performance & Usability Constraints:** In academia, functional correctness is often prioritized over user experience. In a production POS environment, interface layout, visual hierarchy, button placement, and execution speed directly impact retail cashier productivity and customer satisfaction.
- **Collaborative Codebases & State Management:** Academic work is frequently completed individually. In industry, frontend developers must strictly align template tags, form fields, and JavaScript variables with backend team models and Git repository standards.

1.6 Challenges and Areas for Growth

1.6.1 Challenges

During the development of the POS interface, several technical and operational challenges were encountered:

- **Designing an Uncluttered Checkout Screen:** The ‘New Sale’ page required fitting product search results, dynamic cart items, pricing summaries, customer lookups, and payment controls onto a single screen without overwhelming the user.
- **Real-Time Client-Side Calculations:** Synchronizing **JavaScript** cart calculations (subtotal, product discounts, customer discounts, 5% tax) with backend validation logic to prevent calculation discrepancies.
- **Responsive Layout Consistency:** Ensuring complex tables, dashboards, and modal dialogs rendered correctly on both high-resolution monitors and smaller counter touchscreens.
- **Time Management:** Balancing rapid template creation with thorough **UI/UX** testing and documentation within the 5-week schedule.

1.6.2 Areas for Growth

This experience identified key technical areas for continued professional development:

- **Advanced client-side JavaScript frameworks** (e.g., **React.js** or **Vue.js**) for single-page application (SPA) architectures.
- **Advanced UI/UX wireframing** and interactive prototyping techniques.
- **Systematic cross-browser automated UI testing** and frontend performance profiling.

1.7 Industrial Training Details Plan

1.7.1 Training Resource Plan

The training was conducted with structured mentorship and technical infrastructure:

- **Human Resources:** Guidance from Industrial Supervisor **Nusrat Jahan (Data Scientist, O'Dell Tech Ltd.)**, mentorship from development team members, and academic supervision from **Ms. Mayeesha Farjana (Lecturer, Dept. of CSE, Green University of Bangladesh)**.
- **Technical Resources:** Development workstation, **Visual Studio Code IDE**, Web Browsers (**Chrome DevTools**), **HTML5, CSS3, JavaScript, Bootstrap 5, Git/GitHub, XAMPP** local environment, and hosting on **cPanel/subdomain** platforms.

1.7.2 Training Time Management Plan

The industrial training followed a structured timeline spanning CV submission, onboarding, **34 days** of intensive workplace training (**16 August to 19 September 2026**), report drafting, and defense preparation.

The overall planning sequence of the industrial placement is illustrated in **Figure 1.1**.

Industrial Training Timeline & Gantt Schedule												
O'dell Tech Ltd. Point of Sale (POS) Web System Project												
SL	Activity / Stage	Est. Start	Est. End	Duration	Status	Timeline Period Tracker (2026)						
						Jul (1-15 Jul)	½ Jul (16-31 Jul)	Aug (1-15 Aug)	½ Aug (16-31 Aug)	Sept (1-15 Sept)	½ Sept (16-30 Sept)	Oct (1-15 Oct)
1	Company Search & CV Drop	2026-07-01	2026-07-17	17 Days	Completed	17 Days						
2	Selecting & Collecting Forwarding letter	2026-07-18	2026-07-31	14 Days	Completed		14 Days					
3	Received Offer Letter & join	2026-08-01	2026-08-13	13 Days	Completed			13 Days				
4	Training at Odell Tech	2026-08-16	2026-09-19	34 Days	Completed				34 Days			
5	Project Development & Submission	2026-09-04	2026-09-19	16 Days	Completed					16 Days		
6	Industrial Report Submission	2026-09-20	2026-09-27	8 Days	Completed						8 Days	
7	Preparation for Defence	2026-09-28	2026-10-02	5 Days	Completed						5 Days	
8	Defence and waiting for Grade	2026-10-03	2026-10-15	12 Days	Running							Running
TOTAL DURATION & TRACKING PERIOD				119 Days	On Track							

Figure 1.1: Time Management of the Industrial Training Process

1.8 Risk Analysis and Resolve Process

Developing a commercial web interface carries inherent technical and workflow risks. Potential risks identified during the internship included:

- **Unclear Interface Requirements:** Resolved through early mockup discussions and **BRS reviews** with supervisors.
- **Inconsistent UI vs. Backend Data:** Resolved by conducting integrated testing between frontend forms and **Django views**.
- **Cross-Browser Layout Breaking:** Resolved using **Bootstrap grid standards** and multi-browser responsive testing.

The development work followed a systematic multi-stage resolution process from requirement gathering to frontend execution, bug fixing, QA, and deployment as depicted in **Figure 1.2**.

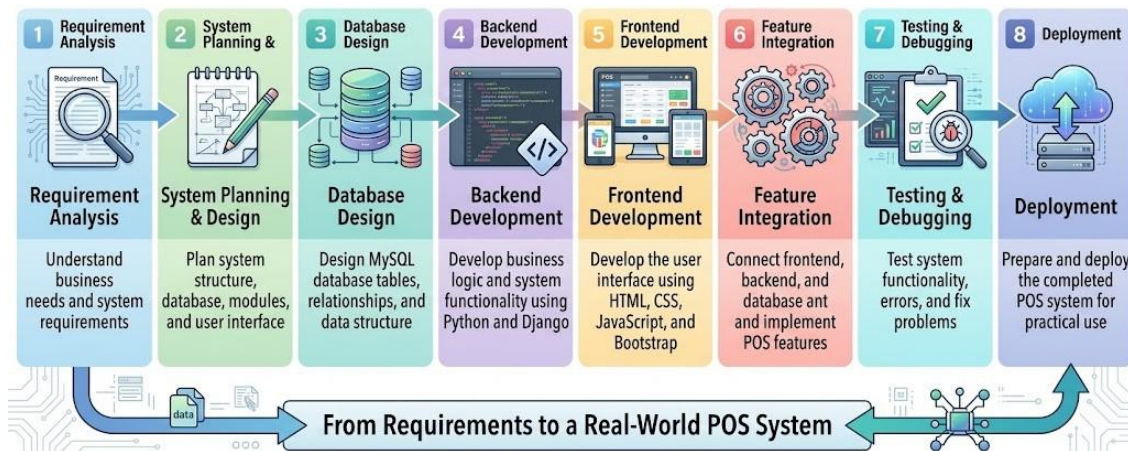


Figure 1.2: Four Steps of the Risk Management Process

1.9 Industrial Training Report Outline

This report is organized into five main chapters and supporting annexures:

- **Chapter 1: Introduction** - Outlines the problem domain, objectives, scope, academic gaps, training plan, and risk management.

- **Chapter 2: Organization Overview** - Describes **O'Dell Tech Ltd.**, its corporate profile, work culture, and functional organizational structure.
- **Chapter 3: Training Activities and POS System Development** - Details weekly activities, frontend engineering tasks, POS interface modules, testing, QA, and deployment.
- **Chapter 4: Acquired Skills During Industrial Training** - Reflects on technical mastery, UI/UX learning, soft skills, teamwork, and professional ethics.
- **Chapter 5: Conclusion** - Summarizes overall achievements, challenges, study scope, and **Program Outcome (PO)** mapping.
- **Annexure & References** - Includes **Program Outcome (PO)** mappings, **Sustainable Development Goal (SDG)** mappings, and formal references.

1.10 Conclusion

Chapter 1 established the foundation of my industrial training report by defining the problem domain, motivation, technical scope, and structural plan of my internship at **O'Dell Tech Ltd.** The hands-on experience in frontend engineering directly enhanced my technical competencies and provided practical context to my academic studies. The following chapter provides an in-depth overview of the host organization and its development environment.

Chapter 2

Organization Overview

2.1 Introduction

O'Dell Tech Ltd. is a modern software solutions and technology firm operating as a sister concern of the **O'Dell Group**. The organization specializes in delivering enterprise digital solutions, with a primary focus on technology integration for Bangladesh's **Ready-Made Garments (RMG)** sector and commercial retail enterprises. Through its specialized software development division and the **O'Dell Tech Innovation Hub**, the company supports project-based software engineering, technological experimentation, and practical industrial training for aspiring computer science professionals.

During my industrial internship, I was placed in the software engineering department, where I worked on the **user interface design, frontend implementation, and client-side integration** of the web-based **Point of Sale (POS) Management System** under industrial supervision.

2.2 Company Profile

O'Dell Tech describes itself as a technology company dedicated to solving critical challenges faced by Bangladesh's RMG factories. The company's stated objective is to help businesses move toward more efficient, technology-driven, and data-oriented operations. The company focuses on providing solutions that can improve operational efficiency, compliance, security, integration, and productivity. Several benefits of its solutions, including streamlined HR processes, social-audit

compliance support, reliable IT infrastructure, software integration, and future-ready technological capabilities.

O'Dell Tech's main solution areas include:

- Human Resources Management Solution
- Merchandising and Production Management Solution
- Production Sample Management Solution
- IT-enabled Solutions
- Web Design and Development
- Artificial Intelligence and Machine Learning
- Server Management
- Software, Applications and Quality Assurance

The company's listed office address is House 1155, Road 10, Avenue 11, Mirpur DOHS, Dhaka.

2.2.1 Historical Timeline

The major stages in the development of O'Dell Tech are presented in Figure 2.1. The timeline highlights the organization's development from its early focus on technology solutions for the Ready-Made Garments (RMG) industry to its expansion into digitalization, software development, Artificial Intelligence (AI), Machine Learning (ML), and professional learning initiatives.

- a. **Early Development:** O'Dell Tech was established as a sister concern of O'Dell Group with a focus on providing technology solutions for the Ready-Made Garments (RMG) industry.
- b. **2023 – RMG Digitalization:** O'Dell Tech strengthened its focus on the digitalization and automation of the RMG sector, particularly in Human Resource Management, compliance, and data-driven decision-making.

- c. **2024 – Expansion of Technology Services:** The organization expanded its technology services to include software development, web development, Artificial Intelligence and Machine Learning, server management, software quality assurance, and production and merchandising management solutions.
- d. **2025 – Innovation and Learning Platform:** O’Dell Tech developed the O’Dell Tech Innovation Hub as a learning platform offering practical technology education, industrial attachment, internships, training, and skill-development opportunities.
- e. **2026 – Integrated and AI-Driven Solutions:** O’Dell Tech continues to develop integrated technology solutions for the RMG sector, with emphasis on automation, ERP-oriented solutions, Artificial Intelligence, operational efficiency, and data-driven business management.

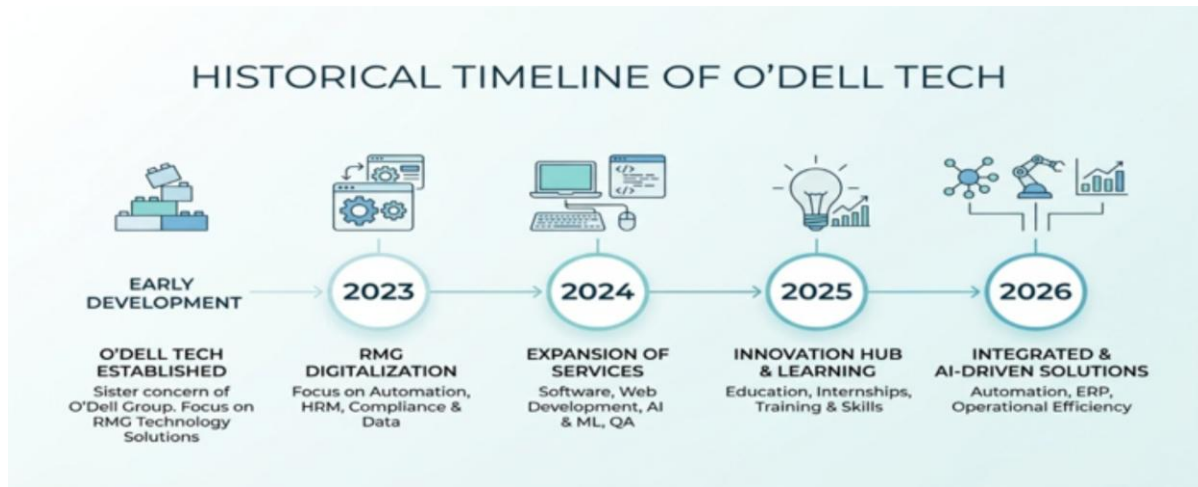


Figure 2.1: Historical Timeline of O’Dell Tech

2.2.2 Work Environment

The work environment at O’Dell Tech provided an opportunity to work with practical software development activities and industry-oriented projects. The company also operates an Innovation Hub that focuses on practical training, industrial attachment, courses, workshops, and research-oriented activities. The Innovation Hub emphasizes learning through practical projects and real-world industry practices. Its programs include internship programs, industrial attachment, up-

skilling courses, workshops, and a research hub. During my internship, I worked under supervision and participated in software development activities related to the POS project. The working environment required communication with supervisors and team members, understanding project requirements, implementing features, testing functionality, and resolving technical issues

2.2.3 Mission and Vision

According to O'Dell Tech's official information, the company is focused on using technology to address challenges in the RMG sector and support businesses in becoming more efficient, compliant, integrated, and technologically advanced.

Mission

- **Company Mission:** To provide technology-driven solutions that improve business operations and contribute to sustainable growth.
- **Innovation Hub Mission:** To create a learning ecosystem where aspiring professionals can develop technical expertise through practical experience, mentorship, and industry-oriented training.

Vision

- **Innovation Hub Vision:** To become an innovation platform that promotes creativity, technological excellence, collaboration, continuous learning, and sustainable growth.

2.2.4 Core Values

The official company information emphasizes several principles in its approach to technology and business solutions. These include:

- Innovation and technological advancement
- Operational efficiency
- Compliance and reliability

- Integration and visibility
- Customer-oriented solutions
- Continuous improvement
- Practical learning and professional development

The Innovation Hub also emphasizes creativity, collaboration, continuous learning, practical experience, and mentorship as important elements of its learning environment.

2.3 Organizational Structure



Figure 2.2: O'Dell Tech Organizational Chart

2.3.1 Roles and Responsibilities of Officials

Based on the organizational structure of O'Dell Tech Ltd. shown in Figure 2.2, the major positions and their respective functional responsibilities can be summarized as follows:

- **Chairman:** Provides overall leadership and strategic direction for the organization.
- **Vice Chairman:** Supports the Chairman in organizational leadership, management, and strategic decision-making.

- **Managing Director:** Oversees the overall operations of the organization and coordinates the major functional departments.
- **HR & Payroll:** Manages human-resource-related activities, employee records, payroll, attendance, and other administrative functions.
- **Software Developer:** Responsible for software development, application implementation, system maintenance, and technical problem-solving.
 - **Senior:** Handles advanced development tasks, technical guidance, code review, and supports junior developers.
 - **Junior:** Assists in software development, coding, debugging, testing, and implementation tasks under supervision.
- **QA & Support:** Responsible for software quality assurance, system testing, issue identification, user support, and problem resolution.
 - **Senior:** Coordinates testing activities, reviews software quality, and provides technical support and guidance.
 - **Junior:** Performs testing, identifies issues, documents problems, and assists in support activities.
- **Marketing:** Responsible for promoting the organization's products and services and supporting business development activities.
 - **Senior Digital Marketer:** Handles digital marketing activities, online promotion, content-related activities, and digital communication.
- **Research Center & Learning Platform:** Supports research, professional development, technical learning, and knowledge-based activities within the organization.
 - **Innovation Hub:** Provides practical training, mentorship, workshops, and industry-oriented learning opportunities. The Innovation Hub connects learners with experienced professionals and provides practical experience through real-world projects.

- **Industrial Supervisor:** Provides guidance to interns regarding assigned activities, technical progress, requirements, implementation, testing, and problem resolution.

2.3.2 Collaboration and Interdepartmental Coordination

Interdepartmental communication was critical during the POS system development . Frontend template development required constant alignment with backend database models and view functions . Regular sync sessions were held to verify that input forms, search bars, cart models, and dynamic invoice views accurately exchanged data with the backend.

2.4 Products/Services

O'Dell Tech provides a range of technology solutions and services.

2.4.1 Human Resources Management

O'Dell Tech provides a cloud-based HRM solution for RMG factories. The system covers employee management, attendance, leave management, employee movement, access privileges, payroll, and reporting. The company also highlights compliance with labor-law requirements and social-audit requirements.

2.4.2 Merchandising and Production Management

The company provides solutions for merchandising and production management in the RMG sector. These solutions are intended to improve workflow efficiency, provide better visibility into production processes, support informed decision-making, and help organizations manage production activities.

2.4.3 Production Sample Management

O'Dell Tech lists a Production Sample Management Solution designed to support sample-related processes. The company states that its solutions are intended to provide better integration and

visibility across business operations and incorporate future AI-enabled capabilities.

2.4.4 Web Design and Development

O'Dell Tech provides web design and development services for creating digital solutions and on-line platforms.

2.4.5 Artificial Intelligence and Machine Learning

The company lists Artificial Intelligence (AI) as one of its technology service areas and describes its AI and Machine Learning capabilities as part of its approach to transforming business data into useful intelligence.

2.4.6 Server Management

O'Dell Tech also provides server management services intended to support reliable and secure IT infrastructure.

2.4.7 Software, Applications and Quality Assurance

The company provides software and application development along with Quality Assurance services. Its website describes software development and QA as important parts of its technology service portfolio.

2.4.8 Innovation Hub and Training

The O'Dell Tech Innovation Hub provides internship programs, industrial attachment, upskilling courses, workshops, and research-related activities. Its training categories include web development, big data and data science, programming languages, databases, AI and machine learning, mobile application development, and UI/UX.

2.5 Clients

O'Dell Tech primarily serves organizations in the Ready-Made Garments (RMG) industry in Bangladesh. The company develops technology solutions to address challenges faced by RMG factories, including productivity management, compliance, HR operations, software integration, and data-driven decision-making.

Several organizations are identified through customer testimonials and case-related statements:

- **Tunic Apparels Ltd:** The company is presented as a customer of O'Dell Tech's HRM software, particularly for attendance tracking, payroll management, and compliance-related activities.
- **Magic Works:** O'Dell Tech presents Magic Works as a customer associated with its IT infrastructure solutions, including security systems, access control, and data backup services.
- **Wraps Clothes Lines:** The organization is presented as a customer using O'Dell Tech's software integration solutions to improve visibility across operational processes and support data-driven decision-making.
- **Yagi Bangladesh Garments LTD:** O'Dell Tech presents Yagi Bangladesh Garments LTD as a customer associated with its AI-enabled and future-ready technology solutions for the RMG sector.

Overall, the publicly identified clients demonstrate O'Dell Tech's focus on providing technology-based solutions to RMG organizations, including HR management, IT infrastructure, software integration, AI-enabled solutions, and operational management.

2.6 Direct Involvement of Resources During Training

During my industrial training, I was directly involved in user interface (UI) design, user experience (UX) enhancement, and frontend implementation for the web-based POS management system. The training provided practical exposure to modern frontend workflows, responsive layout design, client-side scripting, design-to-code implementation, and integration with backend APIs.

The major technical resources and tools used during the training included:

- **Design & Prototyping:** Figma, Wireframing tools
- **Frontend Technologies:** HTML5, CSS3, JavaScript (ES6+), Bootstrap
- **Backend & Template Integration:** Python, Django (Django Template Engine)
- **Database & Local Environment:** MySQL, XAMPP
- **Version Control & Collaboration:** Git, GitHub
- **Development Environment:** Visual Studio Code, Modern Web Browsers (DevTools)

The training began with mastering core frontend fundamentals, UI/UX design principles, and responsive web practices before contributing to the core POS system. Within the POS project, I focused on designing intuitive workflows and responsive user interfaces for product management, customer billing, inventory tracking, checkout/payment counters, return processing, and real-time reporting dashboards.

My responsibilities included translating user requirements into interactive UI layouts, improving usability and accessibility (UX), building responsive frontend components, integrating templates with backend endpoints, conducting cross-browser testing, debugging UI defects, and preparing frontend documentation.

2.7 Constructive Feedback About the Training Program

The industrial training provided practical exposure to software development beyond academic exercises. One of the most important aspects of the training was the opportunity to work with interconnected application modules rather than isolated programming tasks. The training also demonstrated the importance of understanding requirements before implementation. Changes in one module could affect other parts of the application, particularly in areas such as sales, inventory, payments, and reporting. The Innovation Hub's industry-oriented approach is also relevant to this experience because it emphasizes practical learning, real-world projects, mentorship, and bridging academic knowledge with industry requirements. For further improvement, structured code-review sessions, additional documentation of development standards, and more formalized testing reviews could provide interns with deeper exposure to professional software engineering practices.

2.8 Conclusion

O'Dell Tech is a technology company and a sister concern of O'Dell Group that focuses strongly on digital solutions for Bangladesh's Ready-Made Garments industry. Its public service portfolio includes HR management, merchandising and production management, production sample management, web development, AI, server management, software applications, and QA.

The company's Innovation Hub extends this technology-focused environment into practical education through internships, industrial attachments, courses, workshops, and research-oriented activities.

My industrial training at O'Dell Tech provided practical experience in software development and allowed me to apply academic knowledge to a real web-based POS project. The organizational environment, supervision, technical resources, and project-based activities contributed significantly to my understanding of professional software development.

Chapter 3

Specific Details on Training Activities and Report

3.1 Introduction

My industrial internship at **O'Dell Tech Ltd.** took place from **16 August 2026 to 19 September 2026**, accumulating a recorded total of **139 hours and 04 minutes**. The placement was structured to provide practical immersion in commercial web development workflows under the supervision of industrial mentor **Nusrat Jahan** (Data Scientist, O'Dell Tech Ltd.) and academic mentor **Ms. Mayeesha Farjana** (Lecturer, Department of CSE, Green University of Bangladesh).

The core technical assignment was contributing as a **Frontend Developer** and **UI/UX Designer** to a web-based **Point of Sale (POS) Management System** for retail enterprise operations. Initial activities included workspace orientation, Business Requirement Specification (**BRS**) discussions, and a practice task-management web application. The primary assignment involved designing, implementing, and optimizing responsive user interfaces across all major POS modules. This chapter details my weekly duties, technical project contributions, interface design workflows, UI testing methodologies, deployment verification, and internship performance indicators.

3.2 Weekly Duties and Activities

3.2.1 First Week: Orientation, Requirement Review, and UI Planning (16–20 August 2026)

During the first week, I was introduced to O’Dell Tech Ltd.’s software engineering department, workplace policies, team members, and development tools.

- **BRS Review & Requirement Analysis:** Participated in requirement breakdown sessions to analyze the Business Requirement Specification (BRS) document for the POS system. Reviewed user roles (Administrator, Manager, Cashier, Inventory Staff) and visual layout needs.
- **UI/UX Wireframing:** Sketched preliminary screen layouts and visual workflows for the primary sales checkout screen, product catalog tables, and dashboard summary views.
- **Architecture Overview:** Studied how Django’s Model-View-Template (MVT) architecture handles static CSS/JS files, template inheritance (base.html), and view context rendering.
- **Development Process Overview:** Figure 3.1 illustrates the sequential development workflow followed throughout the project.

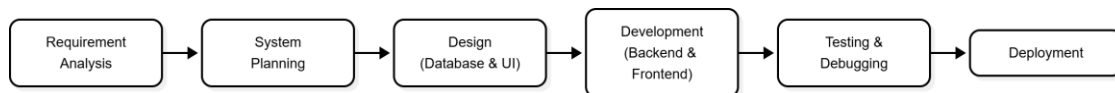


Figure 3.1: SDLC Stages

3.2.2 Second Week: Django Templating and Practice Application (23–27 August 2026)

The second week focused on practical frontend-backend integration exercises to prepare for the main POS system.

- **Environment Setup:** Configured Visual Studio Code, Git, GitHub repository access, and local XAMPP web server environments.

- **Task Management Practice App:** Designed and styled a practice task-management web application using HTML5, CSS3, Bootstrap 5, and Django template tags.
- **Key Learning Outcomes:** Mastered dynamic form handling, template filters, error messaging displays, and CSRF token placements in HTML forms, as summarized in Table 3.1.

Table 3.1: Django Templating and Practice Activities

Activity	Frontend Work Completed	Learning Outcome
Template Inheritance Setup	Created a reusable base.html layout with navigation sidebars and headers.	Code reusability and visual consistency across pages.
Bootstrap 5 Integration	Applied responsive Bootstrap grid system, flexbox containers, and form controls.	Responsive web layout across desktop and tablet screen sizes.
Dynamic Data Display	Rendered backend task lists dynamically using Django template loops (<code>{% for %}</code>).	Seamless frontend-backend data binding.
Form Styling & Validation	Styled input forms, validation alert messages, and submit buttons.	Improved user experience and client-side usability.

3.2.3 Third Week: POS Interface Design and Module Layouts (30 August – 2 September 2026)

During the third week, intensive UI/UX design and template development began for the primary POS system.

- **Dashboard Layout Design:** Built the central dashboard interface featuring summary cards for daily sales, low-stock warnings, and transaction metrics.
- **Checkout & Sales Screen (New Sale):** Designed the core ‘New Sale’ checkout page, dividing the screen into a dual-column layout product search on the left and dynamic cart summary on the right.

- **Product Management Templates:** Developed product list tables, category filter dropdowns, and single/bulk product entry forms.

The primary functional module groupings designed during this phase are outlined in Figure 3.2.

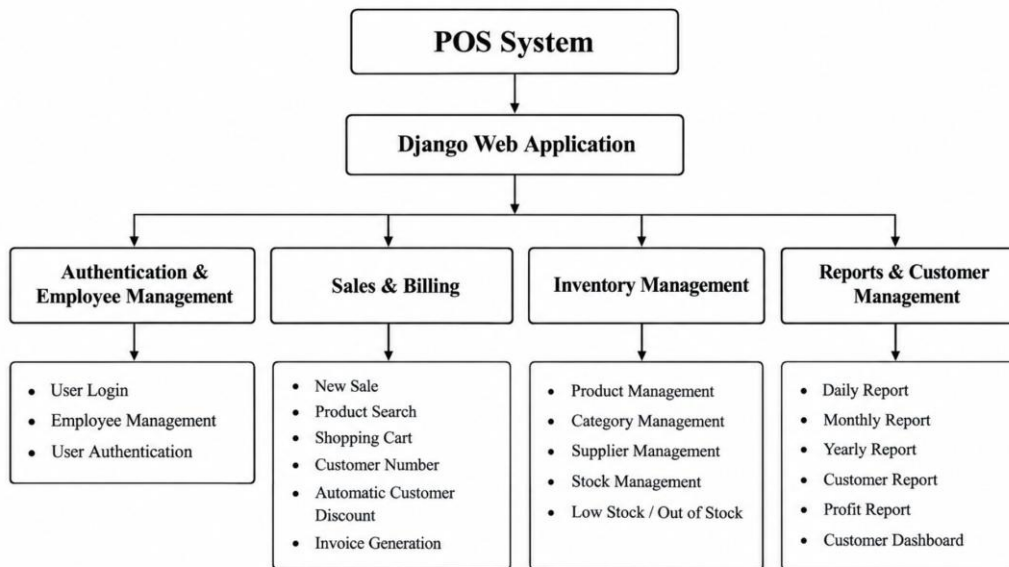


Figure 3.2: Main functional groups of the POS system

3.2.4 Fourth Week: Advanced UI Engineering, Cart Logic, and Testing (8–13 September 2026)

The fourth week centered on client-side dynamic scripting, form handling, and cross-browser visual testing.

- **Client-Side Interactivity (JavaScript):** Programmed vanilla JavaScript functions to enable real-time product search filtering, instant quantity adjustments in the cart, live subtotal updates, and dynamic discount/tax calculations.
- **Printable Invoice Formatting:** Engineered print-specific CSS stylesheets (@media print) for sales receipts, ensuring navigation panels and buttons hide cleanly when printing invoices.

- **User Management Interfaces:** Designed Customer and Employee management table views with status badges, search bars, and action modals.

3.2.5 Fifth Week: Deployment Verification, QA, and Final Report (14–19 September 2026)

The final week focused on quality assurance, hosting checks, and project presentation.

- **Subdomain Deployment Check:** Verified frontend rendering, CSS static file loading, and layout responsiveness on the production subdomain pos.odelltech.com.
- **UI/UX Quality Assurance:** Conducted cross-browser visual testing across Google Chrome and Mozilla Firefox, fixing column alignment bugs and dropdown overlap issues.
- **Final Presentation & Documentation:** Completed report drafting, reviewed deliverables with supervisors, and presented the POS user interface to the O'Dell Tech software team.

3.3 Key Projects

3.3.1 POS Project Scope and My Specific Contributions

The primary project was a web-based Point of Sale (POS) Management System designed to streamline retail shop floor operations. My role focused specifically on Website Design, UI/UX Engineering, Responsive Layouts, and Client-Side Scripting.

My core technical contributions included:

- Designing intuitive, modern wireframes and converting them into clean HTML5/CSS3 templates.
- Utilizing Bootstrap 5 to ensure full responsiveness across various screen resolutions.
- Writing client-side JavaScript for live cart updates, instant product filtering, and price calculations.
- Crafting custom print stylesheets for sales invoices and receipts.

- Collaborating via Git/GitHub to connect HTML templates with Django backend views.

3.3.2 Technologies and Development Tools

The tools and technologies used for frontend development during the internship included:

- **HTML5:** Semantic page structuring, web forms, and template tags.
- **CSS3 & Bootstrap 5:** Custom styling, flexbox grids, modal dialogs, utility classes, and @media print rules.
- **JavaScript (Vanilla JS):** DOM manipulation, client-side event listeners, and dynamic calculations.
- **Visual Studio Code:** Primary IDE for frontend code editing and extension formatting.
- **Chrome DevTools:** DOM inspection, CSS debugging, console log monitoring, and mobile device emulation.
- **Git & GitHub:** Branch management, code commits, and team template synchronization.

3.3.3 System Organization and UI Development Sequence

The interface development followed an iterative design sequence moving from BRS requirements to wireframing, HTML template structure, Bootstrap grid styling, JavaScript interactivity, and cross-browser testing as shown in Figure 3.3.

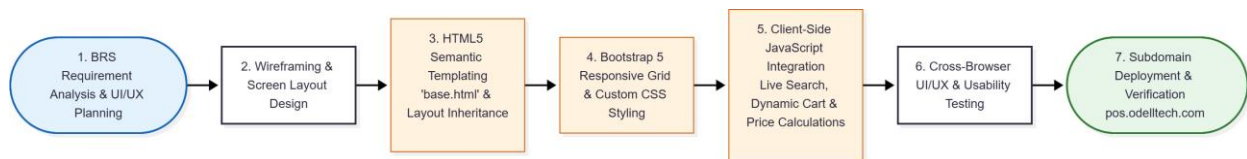


Figure 3.3: Detailed POS development sequence

3.3.4 User Interface Implementation Details

1. POS System Login Interface The login interface (Figure 3.4) features a modern split-screen design. The left panel showcases O'Dell Tech Shopping branding and feature highlights, while the right panel provides clean input fields for user credentials and role-based login guidance.

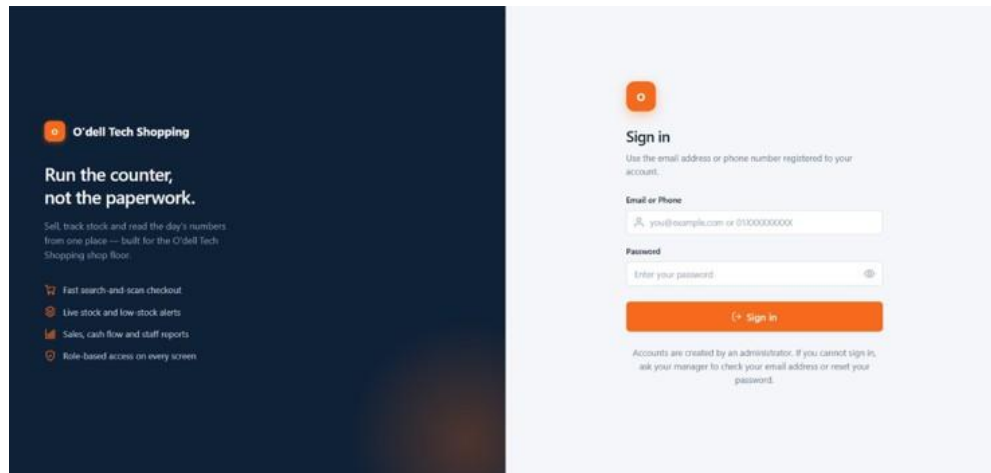


Figure 3.4: POS System Login Interface

2. POS Dashboard Interface The system dashboard (Figure 3.5) provides an executive overview of retail operations. Built using Bootstrap card components, it displays daily sales figures, active transaction counts, total product quantities, low-stock warnings (Yellow), and out-of-stock alerts (Red).

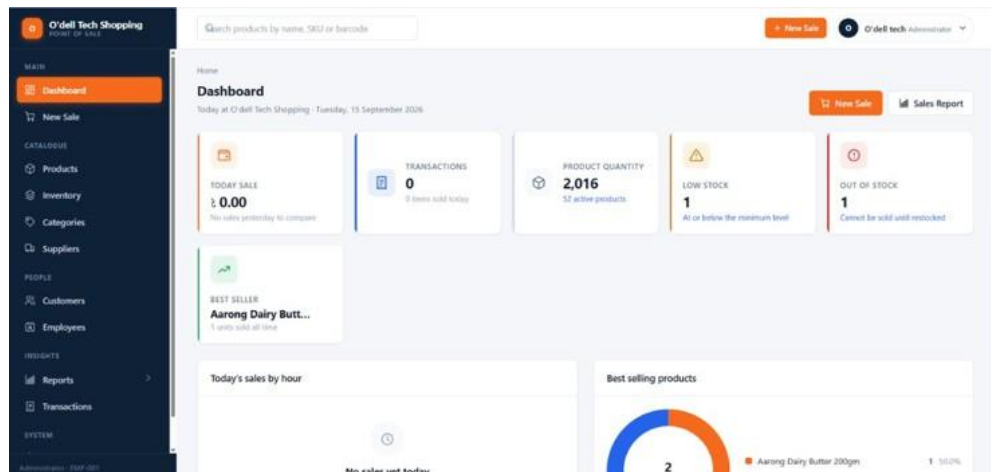
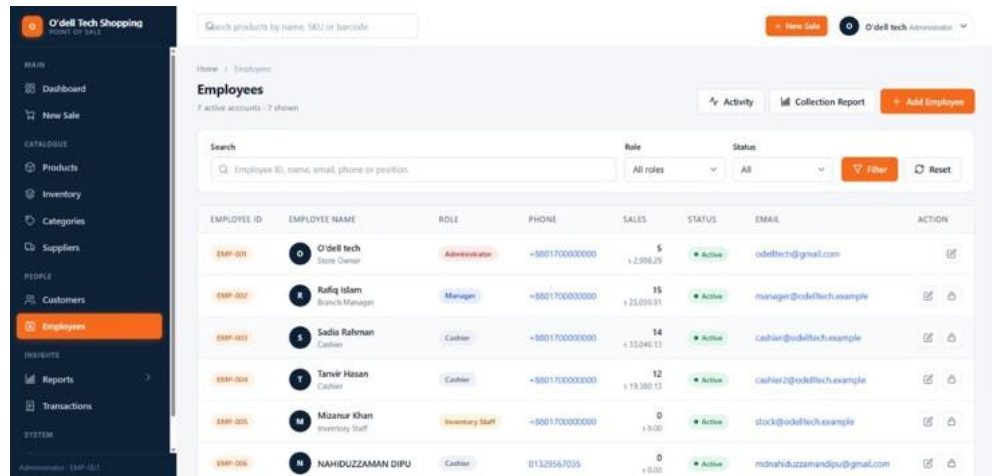


Figure 3.5: POS System Dashboard

3. Employee Management Interface The employee interface (Figure 3.6) displays staff accounts in a clean table format with color-coded role badges (Red for Admin, Blue for Manager, Green for Cashier, Yellow for Inventory Staff) and modal controls for adding or editing employee details.



The screenshot shows the 'Employees' management page. It includes a sidebar with navigation options like Dashboard, New Sale, Products, Inventory, Categories, Suppliers, Customers, Employees (selected), Reports, Transactions, and System. The main content area has a search bar and filters for Role and Status. Below is a table of employees:

EMPLOYEE ID	EMPLOYEE NAME	ROLE	PHONE	SALES	STATUS	EMAIL	ACTION
EMP-001	O'dell tech Store Owner	Administrator	+8801700000000	5 +2,398.29	Active	odelltech@gmail.com	[Edit] [Delete]
EMP-002	Rafiq Islam Branch Manager	Manager	+8801700000000	15 +25,050.91	Active	manager@odelltech.example	[Edit] [Delete]
EMP-003	Sadia Rahman Cashier	Cashier	+8801700000000	14 +33,046.13	Active	cashier@odelltech.example	[Edit] [Delete]
EMP-004	Tanvir Hasan Cashier	Cashier	+8801700000000	12 +19,389.13	Active	cashier2@odelltech.example	[Edit] [Delete]
EMP-005	Mizanur Khan Inventory Staff	Inventory Staff	+8801700000000	0 +0.00	Active	stock@odelltech.example	[Edit] [Delete]
EMP-006	NAHIDUZZAMAN DIPU	Cashier	01325547035	0 +0.00	Active	mdnahiduzzamandipu@gmail.com	[Edit] [Delete]

Figure 3.6: Employee Management Interface

4. New Sale and Interactive Shopping Cart Interface The 'New Sale' checkout screen (Figure 3.7 & Figure 3.8) represents the core operational interface. It features a responsive two-column grid:

- **Left Column:** Instant search bar allowing cashiers to filter products by name, SKU, or barcode, displaying real-time stock availability and unit prices.
- **Right Column (Cart):** Interactive shopping cart displaying selected items, quantity selector buttons (+/-), customer phone number lookup for automatic 2% returning customer discounts, subtotal display, 5% tax calculation, and a prominent 'Complete Sale' action button.

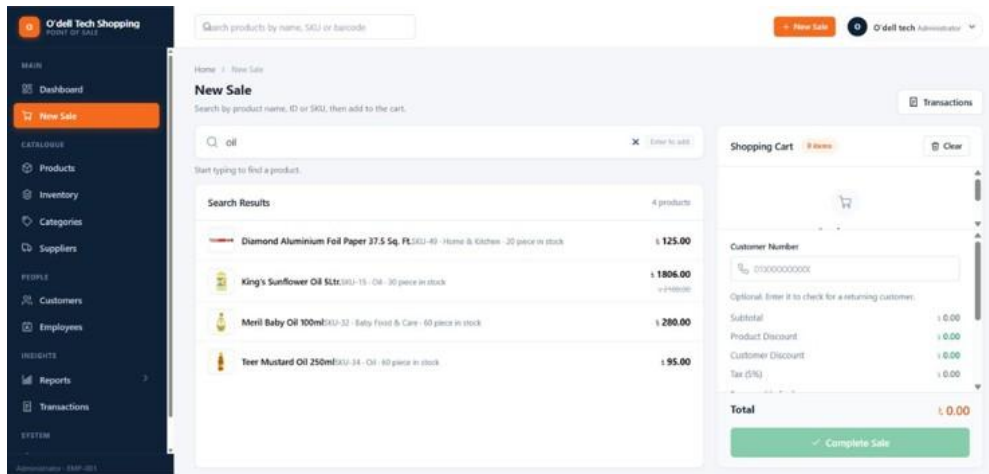


Figure 3.7: New Sale Product Catalog and Search Interface

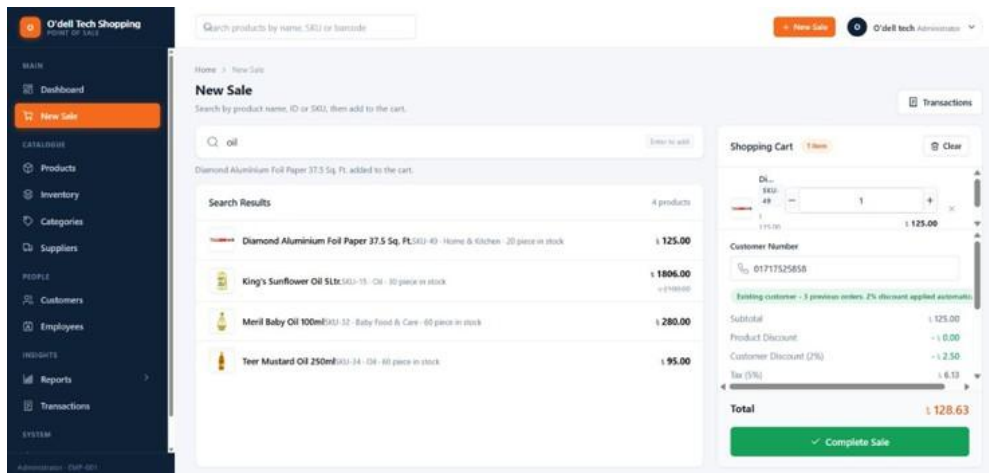


Figure 3.8: Interactive Shopping Cart and Checkout Overview

5. Customer Management Interface The customer interface (Figure 3.9) maintains records of customer phone numbers and purchase history, displaying lifetime shopping totals and discount eligibility badges.

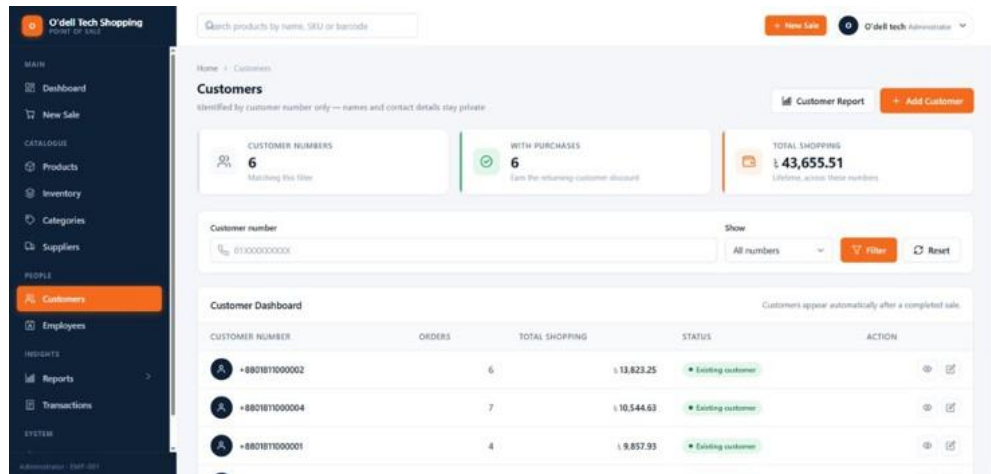


Figure 3.9: Customer Management Interface

6. Printable Sales Invoice Interface The invoice interface (Figure 3.10) displays itemized transaction details, subtotal, customer discounts, tax, total paid, and change returned. Designed with custom @media print CSS rules, clicking the ‘Print’ button outputs a clean paper receipt without web sidebars or navigation headers.

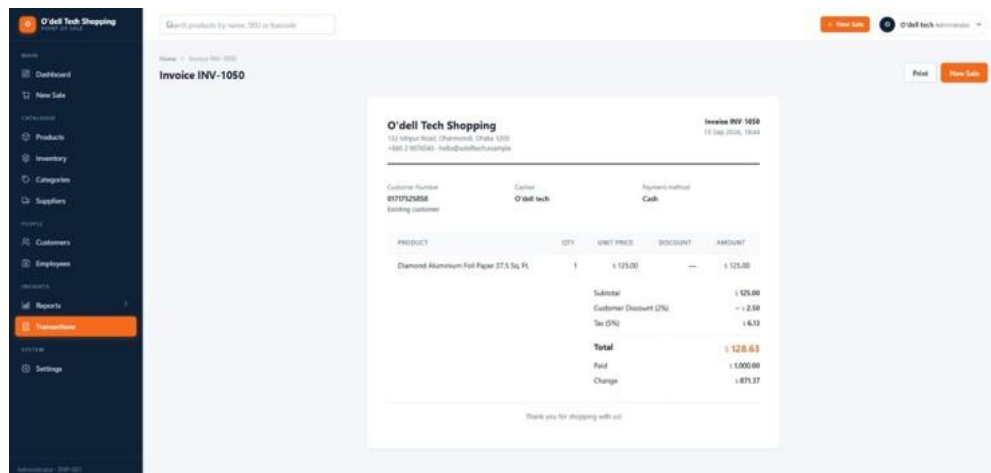


Figure 3.10: Printable Sales Invoice Interface

7. Product and Inventory Management Interfaces The product catalog (Figure 3.11) and bulk upload interface (Figure 3.12) allow staff to filter products by category or stock status, upload Excel catalog files, or manually add individual items through clean modal forms.

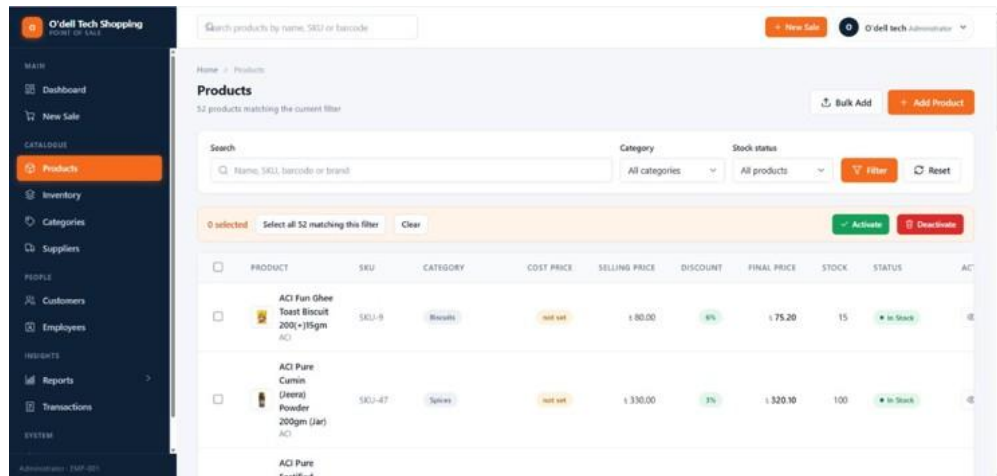


Figure 3.11: Product Management Interface

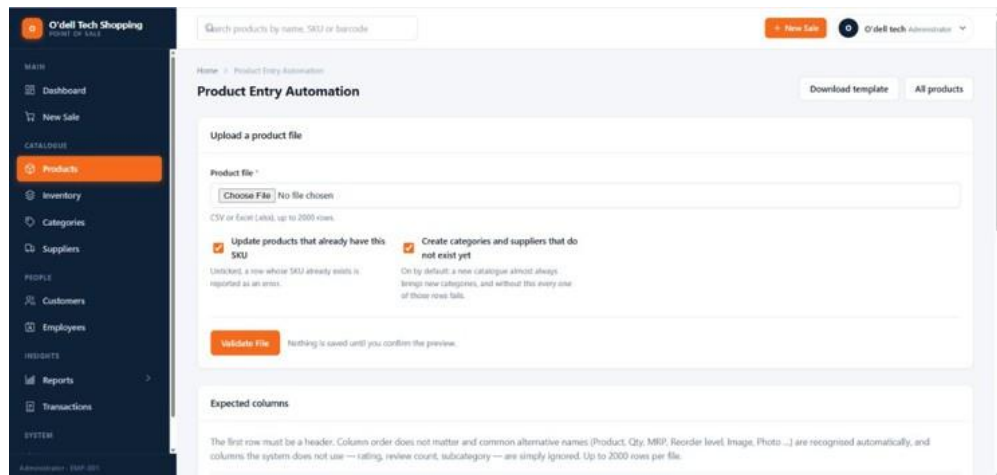


Figure 3.12: Bulk Product Addition Interface

8. Category, Supplier, and Inventory Interfaces Category management (Figure 3.13), Supplier management (Figure 3.14), and Inventory stock tracking (Figure 3.15) provide structured table views for managing product classifications, supplier contact info, and stock value summaries.

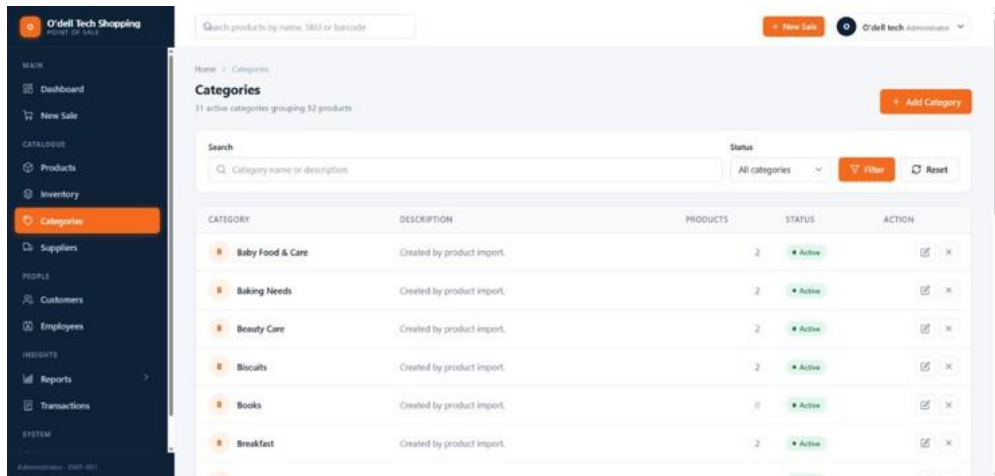


Figure 3.13: Category Management Interface

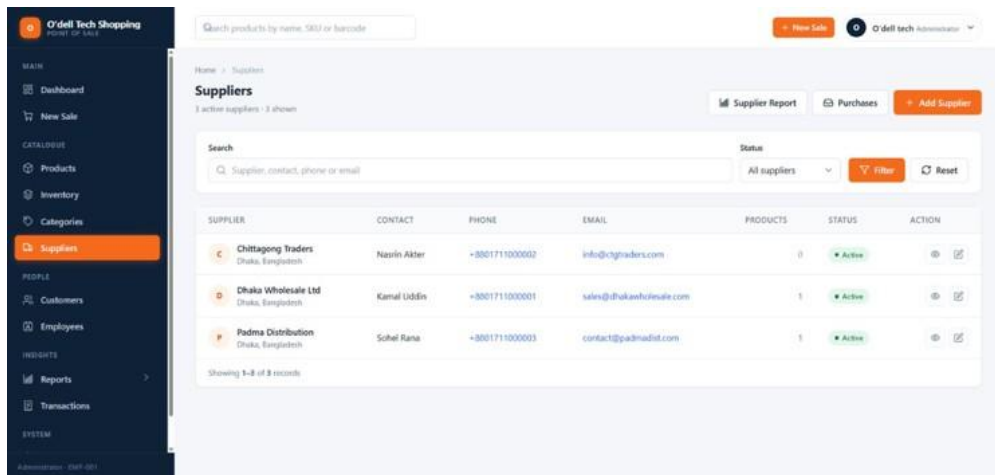


Figure 3.14: Supplier Management Interface

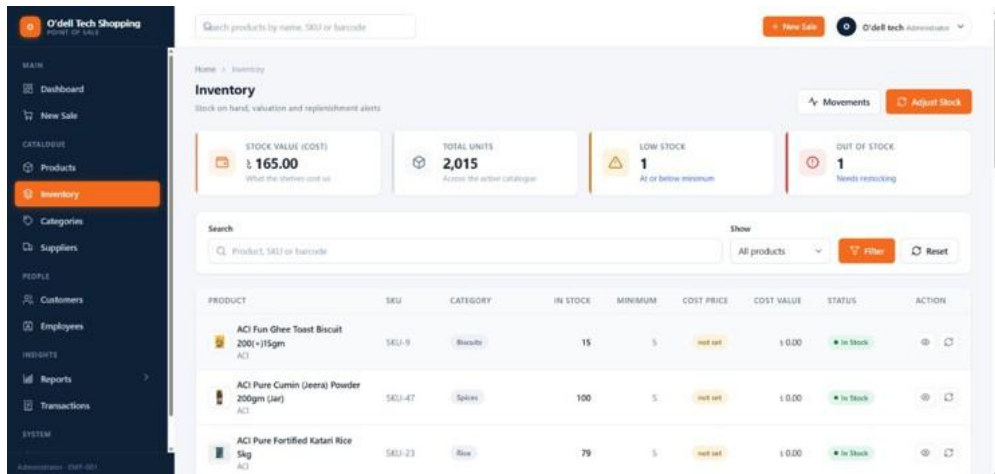


Figure 3.15: Inventory Management Interface

9. Reports, Activity Log, and System Deployment The reporting overview (Figure 3.16 & Figure 3.17) provides visual metric cards and CSV export controls. System operations are logged in the Activity Log (Figure 3.18), while GitHub repository management and subdomain deployment on pos.odelltech.com are captured in Figure 3.19.

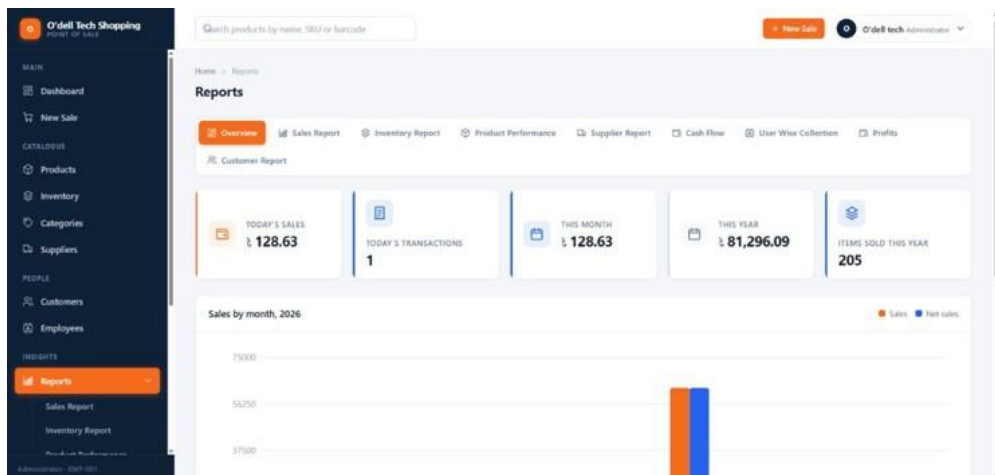


Figure 3.16: Report Overview

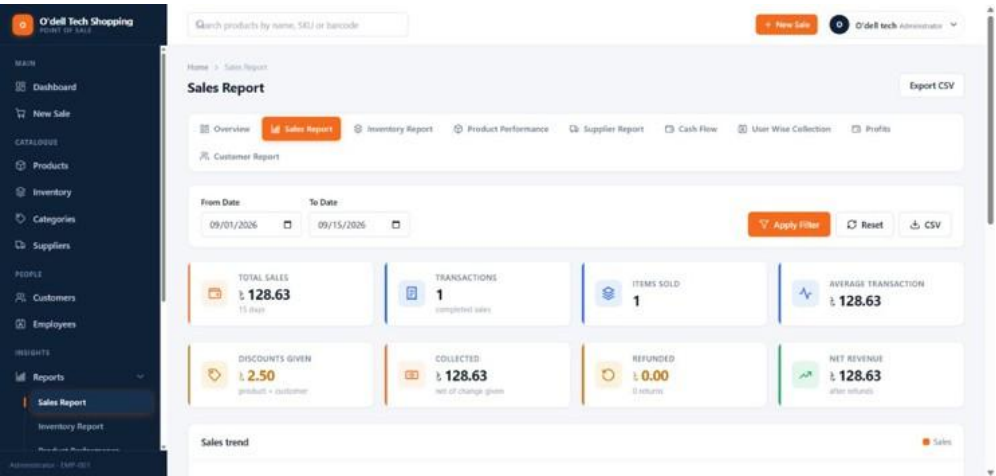


Figure 3.17: Sale Report Overview

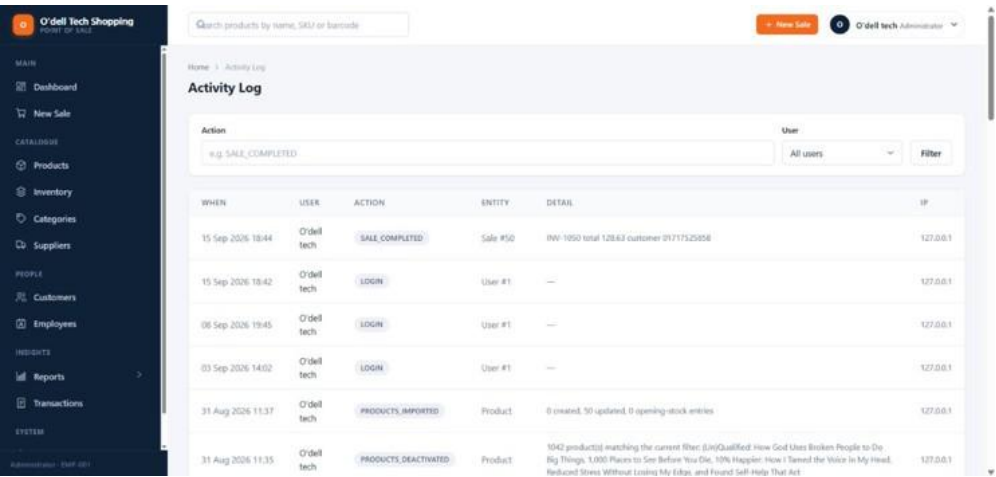


Figure 3.18: User Activity Log

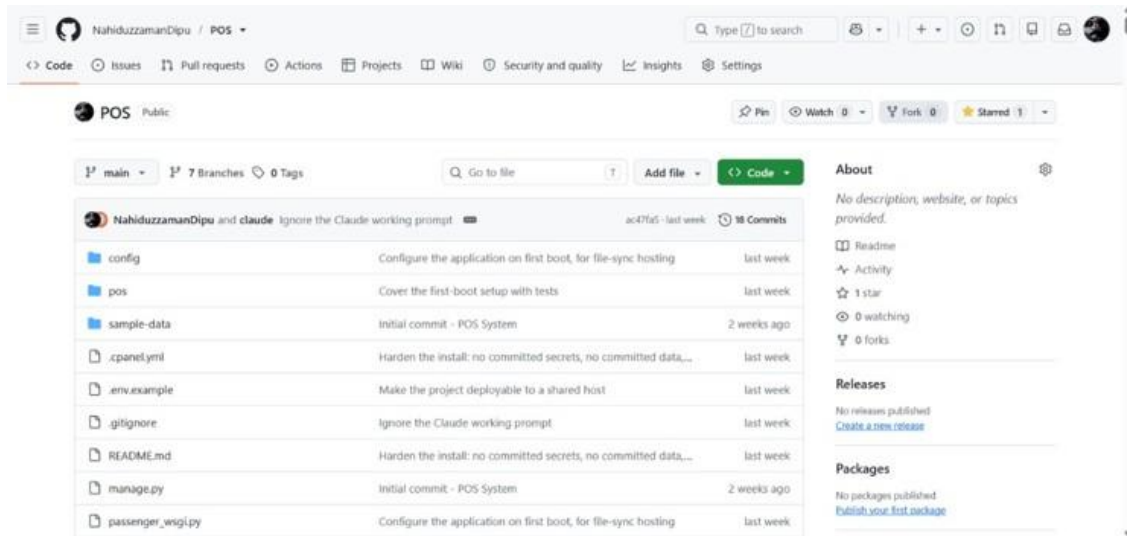


Figure 3.19: POS System GitHub Deployment

3.3.5 Client-Side Testing and UI Validation

Frontend testing was performed to verify layout responsiveness, visual alignment, and client-side scripting accuracy:

- **UI/UX Usability Testing:** Checked form input constraints, ensuring negative quantities or invalid phone numbers trigger visual alert warnings.
- **Cross-Browser Verification:** Tested layout rendering across Google Chrome and Mozilla Firefox.
- **Responsive Emulation:** Verified viewport adaptability on desktop monitors (1920 × 1080) and tablet screen viewports using Chrome DevTools.

3.3.6 Other Responsibilities

In addition to frontend coding, my assigned responsibilities included:

- **BRS Review & Team Sync:** Participating in requirement clarification sessions with supervisors and backend engineers.

- **UI Code Reviews:** Sharing frontend template files on GitHub and incorporating mentor feedback on CSS organization.
- **Report Preparation & Presentation:** Documenting UI workflows and presenting the completed POS user interface to the O’Dell Tech software team.

3.3.7 Key Performance Indicators (KPIs)

3.3.7.1 Project Accomplishment Milestones

My frontend development progress followed strict weekly milestones, as documented in Table 3.2.

Table 3.2: Recorded Development and Testing Milestones

Date	Recorded Frontend Activity
2 September 2026	Completed Product, Category, and Supplier layout templates.
8 September 2026	Completed Customer and Employee management table interfaces.
9 September 2026	Implemented interactive New Sale checkout screen and JavaScript cart logic.
10 September 2026	Designed Inventory stock tracking views and status badge styles.
12 September 2026	Built Reports overview page and printable Invoice CSS layouts.
13 September 2026	Conducted full UI/UX debugging and client-side script corrections.
15 September 2026	Quality Assurance (QA) layout evaluation across browsers.
16 September 2026	Final UI retesting and deployment check on pos.odelltech.com .

3.3.7.2 Recorded Working Hours Evidence

The industrial training working hours were recorded meticulously over five weeks, totaling 139 hours and 04 minutes, as detailed in Table 3.3.

Table 3.3: Weekly Recorded Working Hours

Week	Date Range	Recorded Hours
Week 1	16-20 August 2026	24h 20m
Week 2	23-27 August 2026	17h 5m
Week 3	30 August - 2 September 2026	21h 25m
Week 4	8-13 September 2026	20h 0m
Week 5	14-19 September 2026	18h 45m
Total	16 August - 19 September 2026	103h 35m

3.3.8 Internship Summary

A comprehensive summary of the administrative details, technical scope, and outcomes of my industrial training is presented in Table 3.4.

Table 3.4: Internship Summary

Parameter	Details
Host Organization	O'Dell Tech Ltd.
Organization Location	House 1155, Road 10, Avenue 11, Mirpur DOHS, Dhaka
Training Duration	16 August 2026 to 19 September 2026 (5 Weeks)
Total Recorded Hours	139 hours 04 minutes
Industrial Supervisor	Nusrat Jahan, Data Scientist, O'Dell Tech Ltd.
Academic Supervisor	Ms. Mayeessa Farjana, Lecturer, Dept. of CSE, Green University of Bangladesh
Main Assigned Project	Web-Based Point of Sale (POS) Management System
My Primary Role	Frontend Developer & UI/UX Designer
Technologies Used	HTML5, CSS3, JavaScript, Bootstrap 5, Git, GitHub, VS Code, Chrome DevTools
Core UI Deliverables	Dashboard, New Sale Checkout, Dynamic Cart, Invoices, Product Catalog, Customer/Employee Views
Deployment Platform	pos.odelltech.com

Chapter 4

Acquired Skills During Industrial Training

4.1 Introduction

The industrial training program at **O'Dell Tech Ltd.** from **16 August 2026 to 19 September 2026** served as a transformative bridge connecting my academic CSE coursework at **Green University of Bangladesh** with commercial software development practices. Working on a live, web-based **Point of Sale (POS) Management System** provided invaluable hands-on experience in translating abstract business requirements into responsive, interactive, and user-friendly web interfaces.

While academic projects often focus on backend scripts or isolated static pages, this internship allowed me to master modern frontend engineering workflows, responsive **UI design principles**, client-side dynamic scripting, and collaborative version control within **Django's Model-View-Template (MVT)** architecture. This chapter details the technical competencies, theoretical knowledge, practical insights, soft skills, and professional ethics I developed throughout the training period.

4.2 Learning New Technologies and Applying Them to Solve Engineering Problems

During the internship, I learned, adapted, and applied several industry-standard frontend tools and practices to build and optimize the **POS** user interface:

- **Bootstrap 5 & Responsive Grid Architecture:**

- **Proficiency Level:** Advanced.
- **Application:** Designed and implemented fully responsive screen layouts across all core **POS** modules, including the **Dashboard**, **New Sale interface**, **Interactive Cart**, **Customer/Employee tables**, and printable **Invoices**.
- **Problem Solved:** Resolved screen clutter and layout breakage on smaller display screens by leveraging **Bootstrap's 12-column grid system**, **flexbox utilities**, **responsive navigation sidebars**, and **custom modal components**.

- **HTML5 & Semantic Markup:**

- **Proficiency Level:** Advanced.
- **Application:** Built clean, structured, and accessible **HTML** templates using **Django's template engine** (**Template Inheritance**, `{% block %}` tags, and template filters).
- **Problem Solved:** Replaced repetitive **HTML** code across multiple pages by creating a centralized base template (`base.html`), ensuring visual consistency and easier maintainability across the system.

- **CSS3 & Custom Visual Styling:**

- **Proficiency Level:** Intermediate-to-Advanced.
- **Application:** Wrote custom **CSS** rules, styled cards, badge indicators (e.g., low-stock warnings, active status), and designed print-specific styles (`@media print`) for sales invoices.
- **Problem Solved:** Designed a clean, professional invoice template that automatically hides navigation sidebars, buttons, and footers when printed or saved as a **PDF**.

- **Vanilla JavaScript & DOM Manipulation:**

- **Proficiency Level:** Intermediate.

- **Application:** Implemented dynamic client-side logic for instant product searches, dynamic cart item quantity adjustments, live subtotal calculations, and instant discount/tax updates.
- **Problem Solved:** Minimized unnecessary page reloads during checkout by handling cart row additions and price updates dynamically on the client side before submitting the final sale payload.
- **Git & GitHub for Version Control:**
 - **Proficiency Level:** Intermediate.
 - **Application:** Managed code revisions, created development branches, resolved merge conflicts, and synchronized frontend template files with backend team commits.
 - **Problem Solved:** Prevented code overwrites when integrating frontend **HTML/CSS** changes with backend views and **ORM models** developed by team members.
- **Chrome DevTools & Client-Side Debugging:**
 - **Proficiency Level:** Intermediate.
 - **Application:** Utilized browser console inspection, element styling tweaks, network tab monitoring, and responsive view emulation.
 - **Problem Solved:** Identified and fixed frontend layout alignment issues and **JavaScript** execution errors across **Google Chrome** and **Mozilla Firefox** browsers.

4.3 Theoretical Knowledge Gained

4.3.1 UI/UX Design Principles & Human-Computer Interaction (HCI)

- **Visual Hierarchy and Spatial Organization:** Applied principles of **cognitive load reduction** by grouping related checkout controls logically (**Product Search** on the left, **Cart and Summary** on the right).

- **Feedback and Affordability:** Designed explicit visual cues such as color-coded status badges (**Green for In Stock, Yellow for Low Stock, Red for Out of Stock**), hover effects on action buttons, and **modal confirmation dialogs**.

4.3.2 Software Testing Principles for User Interfaces

- **Usability & Functional Validation:** Understood the distinction between **functional correctness** and **usability testing**. Verified that forms correctly restrict invalid inputs (e.g., preventing negative quantities or non-numeric discount values).
- **Cross-Browser & Multi-Device Standards:** Learned theoretical concepts regarding **responsive viewport standards**, **CSS media queries**, and **cross-browser engine rendering differences**.

4.3.3 Frontend-Backend Coupling in Modern Web Architecture

Understood how **Django's MVT pattern** decouples business logic from presentation layers, allowing frontend developers to work independently on **HTML templates** while binding dynamic data using context variables (`{{ sale.total }}`, `{% for item in cart %}`).

4.4 Practical Knowledge Acquired

4.4.1 Hands-on UI/UX Development Experience

Gained direct experience in **wireframing interface layouts**, converting conceptual requirements into clean **Bootstrap code**, styling table views with pagination controls, and polishing **POS interface elements**.

4.4.2 Defect Tracking and Client-Side Debugging

Actively participated in identifying **UI bugs** such as misaligned table columns, overlapping drop-down menus, and inconsistent font sizes logging observations, implementing **CSS/JS fixes**, and retesting across devices.

4.4.3 Practical Perspective on Starting a Tech Startup

The experience of developing a commercial **POS software** provided practical insights into launching a technology startup focused on retail web solutions:

- **Required Technical Resources:** Mid-to-high performance development workstations, cloud server infrastructure (e.g., **cPanel/AWS/DigitalOcean** hosting), domain setup, and local database servers.
- **Human Resource Plan:** A compact team comprising **1 UI/UX Frontend Engineer, 1 Backend/Database Specialist, 1 QA Tester, and 1 Product Lead/Business Analyst.**
- **Budgetary Considerations:** Software licensing (IDE, hosting, domain, **SSL**), initial hardware overhead, marketing/client onboarding costs, and maintenance reserves.
- **Project Scope for Launch:** Developing a specialized, lightweight, cloud-based **POS system** tailored specifically to local retail shops or boutique clothing stores in **Bangladesh.**

4.5 Ethical Responsibilities and Professional Conduct

During my industrial training at **O'Dell Tech Ltd.**, adhering to professional ethics and industry standards was fundamental to developing a reliable Point of Sale (POS) system. Ethical decision-making directly shaped interface security, transparency in team communication, and accessible layout design. The specific ethical principles applied throughout the internship are mapped to the **IEEE Code of Ethics** as detailed in Table 4.1.

Table 4.1: IEEE Code of Ethics Applied During Industrial Training

IEEE Code	Implementation in Training
Public Safety & Welfare	Ensured user data privacy by designing forms that avoid displaying sensitive customer records unnecessarily and maintaining client-side input validation standards.
Professional Integrity	Maintained complete transparency when reporting UI bugs, technical limitations, and daily sprint progress; avoided masking layout flaws during QA reviews.
Fairness & Inclusion	Designed intuitive visual hierarchies, accessible color contrasts, and clear button labels so that retail cashiers of varying technical skill levels can operate the system equitably.
Technical Competence	Actively researched modern CSS/JS standards, embraced systematic code reviews, and consistently sought supervisor feedback to improve software quality and code efficiency.

4.6 Conclusion

Chapter 4 highlighted the comprehensive skills acquired during my industrial internship at O'dell Tech Ltd.. The practical exposure significantly strengthened my capabilities in frontend engineering, UI/UX design, client-side scripting, and team collaboration. These acquired competencies have provided a solid foundation for my future academic endeavors and professional career in web engineering.

Chapter 5

Conclusion

5.1 Summary

My industrial internship at **O'Dell Tech Ltd.** spanned five weeks from **16 August 2026 to 19 September 2026**, accumulating a total recorded working time of **139 hours and 04 minutes**. The placement provided a structured environment to transition from academic computer science concepts to practical, industry-grade software development practices.

While initial onboarding included skill assessments and a practice task-management application, the primary focus of my training was contributing as a **Frontend Developer** and **UI/UX Designer** to a web-based **Point of Sale (POS) Management System** for retail operations. My responsibilities centered on designing intuitive, responsive visual templates and implementing client-side logic across key functional modules, including the **Dashboard**, **New Sale interface**, **Interactive Cart**, **Customer/Employee management views**, and printable **Invoices**.

This experience reinforced the critical connection between frontend interface design and backend business logic. I gained firsthand experience in creating layout wireframes, writing clean **HTML5/CSS3** templates, implementing dynamic **JavaScript** calculations, ensuring cross-browser responsiveness using **Bootstrap 5**, and collaborating through **Git/GitHub**. The completed POS application was successfully validated and deployed on pos.odelltech.com.

5.2 Challenges During Industrial Training

5.2.1 Understanding Requirements and UI/UX Integration

Designing an efficient **POS interface** required translating complex business rules such as **tiered customer discounts**, **product-level discounts**, and **5% tax rates** into clear, non-cluttered user interface layouts. Ensuring that frontend form fields dynamically reflected backend **Django context data** without breaking visual responsiveness was a major initial challenge.

5.2.2 Client-Side Interactivity and Testing

Implementing live client-side calculations (**instant cart item additions**, **real-time subtotal updates**, and **price adjustments**) required careful **JavaScript DOM manipulation**. Ensuring that client-side calculations strictly matched backend **ORM transaction rules** required rigorous integration checking, debugging, and retesting across various screen viewports.

5.2.3 Time Management and Cross-Team Communication

Balancing initial **UI wireframing**, **template creation**, **CSS styling**, testing, and report drafting within the **5-week schedule** required disciplined time management. Communicating template tag dependencies clearly with backend team members was essential to prevent **merge conflicts** in the shared **GitHub repository**.

5.2.4 Deployment and UI Verification

Transitioning the frontend interface from a local development environment to the live production subdomain (`pos.odelltech.com`) presented layout verification challenges. Conducting follow-up **QA checks** was necessary to ensure **static CSS assets**, **JavaScript files**, and responsive media queries rendered consistently on the hosted server.

5.3 Scope of the Study

This report reflects my direct technical contributions to the user interface design, layout architecture, and client-side interactivity of the web-based POS system . The study focuses on frontend usability, responsive component integration, and UI/UX optimization.

It does not claim direct evaluation of backend database server tuning, non-software business metrics, or long-term financial impacts on O'dell Tech Ltd.. Future enhancements to the interface could explore adopting single-page application (SPA) frontend frameworks like React.js or Vue.js to further accelerate counter checkout speeds.

5.4 Problems and Executions

Throughout the project, specific technical challenges were encountered and systematically resolved:

- **Problem 1: Screen Clutter on Checkout**

- **Issue:** The New Sale page initially contained too many input fields, making product selection difficult for cashiers.
- **Execution:** Redesigned the checkout page layout into a clean two-column grid using **Bootstrap 5** placing instant product search on the left and the dynamic cart summary on the right.

- **Problem 2: Unformatted Invoice Printing**

- **Issue:** Standard web page printing included unwanted navigation bars, sidebars, and operational buttons on customer receipts.
- **Execution:** Created print-specific CSS stylesheets (@media print) that automatically hide non-essential UI elements, producing clean, professional printable receipts.

- **Problem 3: Unnecessary Page Reloads During Checkout**

- **Issue:** Early cart modifications caused full page refreshes, slowing down transaction speed.

- **Execution:** Implemented client-side **JavaScript** event listeners to dynamically modify cart row totals and display real-time price updates prior to form submission.

5.5 Overall Achievement

The primary achievement of this industrial training was the successful delivery, testing, and deployment of a responsive, user-centric web interface for a commercial POS system. The internship significantly elevated my frontend development proficiency, strengthened my understanding of UI/UX design patterns, and built confidence in professional team collaboration.

ANNEXURE 1: GRADUATE PROGRAM OUTCOME (PO) MAPPING

Program Outcomes (POs) represent key benchmarks established by the Department of CSE, Green University of Bangladesh. Table 5.1 details how my industrial training fulfilled these academic outcomes.

Table 5.1: Program Outcome (PO) Fulfillment in Industrial Training

Program Outcome (PO)	Fulfillment Level	Training Knowledge Applied	Evidence / Contribution
PO2: Problem Analysis	Fully Ful-filled	Analyzed checkout UI bottlenecks, misaligned table structures, and client-side pricing calculation errors.	Redesigned checkout layout into a two-column Bootstrap layout and resolved JavaScript calculation bugs.
PO5: Modern Tool Usage	Fully Ful-filled	Utilized modern web technologies including HTML5, CSS3, JavaScript, Bootstrap 5, VS Code, Chrome Dev-Tools, and Git/GitHub.	Developed 10+ responsive template views, tracked revisions via Git, and debugged DOM elements using Dev-Tools.
PO6: The Engineer and Society	Fully Ful-filled	Evaluated accessibility standards and visual clarity for retail cashiers to reduce user error during fast-paced operations.	Designed clear color-coded status badges, intuitive search filters, and accessible modal dialogs.
PO7: Environment & Sustainability	Partially Fulfilled	Designed printable electronic invoices and digital reporting screens to support paperless office practices.	Implemented clean invoice views and CSV export functionality for digital record-keeping.
PO8: Ethics	Fully Ful-filled	Adhered to professional software integrity, IEEE Code of Ethics, and transparent bug reporting.	Accurately documented development progress without masking layout flaws or fabricating metrics.
PO10: Communication	Fully Ful-filled	Communicated UI/UX requirements during daily team stand-ups, prepared technical documentation, and delivered final project presentation.	Presented the POS frontend workflow to the development team and submitted a comprehensive report.
PO12: Life-Long Learning	Fully Ful-filled	Expanded web development knowledge by mastering Bootstrap 5 grid systems and client-side scripting during the internship.	Applied newly learned frontend techniques to build production-ready UI components and identified SPA frameworks for future learning.

Table 5.2: Complex Engineering Problems (WP) – Application to Project

No.	Attribute	Description	How it applies to your problem
Complex problems have characteristic WP1 and some or all of WP2 to WP7.			
WP1	Depth of Knowledge Required	Requires in-depth engineering knowledge of modern frontend frameworks and MVT architecture.	Applied semantic HTML5, custom CSS3, Bootstrap 5 flexbox grids, and JavaScript DOM manipulation to build complex POS views.
WP2	Range of conflicting requirements	Involves balancing user accessibility, visual aesthetics, and dense data displays.	Balanced fitting extensive transaction data (items, prices, tax, discounts, customer history) into a compact, easy-to-read screen layout.
WP3	Depth of analysis required	Requires abstract thinking to structure client-side data flows without server overload.	Formulated client-side JavaScript event handling to calculate subtotals and discounts instantly before final backend submission.

Continued on next page

Table 5.2 – continued from previous page

No.	Attribute	Description	How it applies to your problem
WP4	Familiarity of issues	Involves infrequently encountered rendering issues across diverse client devices.	Resolved cross-browser layout inconsistencies and print formatting issues using CSS media queries (@media print).
WP5	Extent of applicable codes	Adheres to established web standards and UI/UX design guidelines.	Followed W3C HTML/CSS validation standards, Bootstrap grid conventions, and responsive viewport specifications.
WP6	Extent of stakeholder involvement and level of conflicting requirements	Involves addressing the differing needs of Administrators, Managers, Cashiers, and Inventory Staff.	Designed role-specific UI navigation bars and action controls aligned with user authorization levels.
WP7	Interdependence	High-level interface problems involving interconnected UI components.	High-level interface problems involving interconnected UI components.

Table 5.3: Complex Engineering Activities (EA) – Application to Project

No.	Attribute	Description	How it applies to your activity
Complex activities mean (engineering) activities or projects that have some or all of the following characteristics.			
EA1	Range of resources	Involves diverse human, technical, and software tools.	Utilized development workstations, VS Code IDE, Bootstrap 5, Chrome DevTools, Git, GitHub, and local XAMPP web servers.
EA2	Level of interactions	Requires resolving technical conflicts between frontend templates and backend ORM models.	Collaborated closely with backend developers to ensure HTML form field names and CSRF tokens matched Django view specifications.
EA3	Innovation	Involves creative application of frontend engineering principles.	Designed an automated client-side invoice preview and responsive data table filtering system for product catalogs.

Continued on next page

Table 5.3 – continued from previous page

No.	Attribute	Description	How it applies to your activity
EA4	Consequences to society and the environment	Direct impact on operational efficiency and retail user experience.	Reduced cashier input errors and checkout waiting times by delivering a streamlined, highly intuitive POS checkout screen.
EA5	Familiarity	Extending beyond prior academic coursework through practical industry exposure.	Advanced from static HTML assignments to engineering dynamic, production-deployed web templates on live subdomain hosting.

ANNEXURE 2: SUSTAINABLE DEVELOPMENT GOAL (SDG) MAPPING

The United Nations Sustainable Development Goals (SDGs) represent a global framework for sustainable progress. Table 5.4 outlines the alignment of my industrial training with the 17 SDGs.

Table 5.4: Mapping of All 17 SDG Goals with Achievements from Industrial Training

SDG Goal	Relevant Achievements During Training	Fulfillment Level
SDG 1: No Poverty	No direct contribution observed within software development scope.	Not Fulfilled
SDG 2: Zero Hunger	Not directly applicable to retail POS web development.	Not Fulfilled
SDG 3: Good Health & Well-Being	Followed ergonomic workstation practices and maintained healthy work-life balance during training.	Partially Fulfilled
SDG 4: Quality Education	Gained advanced technical skills in frontend engineering, Bootstrap, JavaScript, and Git through industry mentorship.	Fully Fulfilled

SDG Goal	Relevant Achievements During Training	Fulfillment Level
SDG 5: Gender Equality	Worked in an inclusive engineering workplace under female industrial leadership (Nusrat Jahan).	Fully Fulfilled
SDG 6: Clean Water & Sanitation	No direct involvement in this sector.	Not Fulfilled
SDG 7: Affordable & Clean Energy	Not addressed in assigned software tasks.	Not Fulfilled
SDG 8: Decent Work & Economic Growth	Contributed to commercial retail software automation, developing industry-relevant software engineering competencies.	Fully Fulfilled
SDG 9: Industry, Innovation & Infrastructure	Designed modern digital web interfaces to upgrade traditional retail management workflows.	Fully Fulfilled
SDG 10: Reduced Inequalities	Designed clean, intuitive layouts ensuring system accessibility for cashiers of varying technical backgrounds.	Partially Fulfilled
SDG 11: Sustainable Cities & Communities	Supported local business digitalization through cloud-deployable retail management software.	Partially Fulfilled
SDG 12: Responsible Consumption & Production	Created digital invoice views and reporting modules to reduce physical paper usage in retail stores.	Moderately Fulfilled
SDG 13: Climate Action	No environmental energy metrics evaluated in project.	Not Fulfilled
SDG 14: Life Below Water	Not relevant to retail software scope.	Not Fulfilled
SDG 15: Life on Land	No environmental land sustainability components involved.	Not Fulfilled

SDG Goal	Relevant Achievements During Training	Fulfillment Level
SDG 16: Peace, Justice & Strong Institutions	Adhered to IEEE ethical code, maintain transparent progress logging, and implemented role-based user access controls.	Fully Fulfilled
SDG 17: Partnerships for the Goals	Collaborated in cross-functional software teams bridging academic university learning with industrial application.	Fully Fulfilled

References

- [1] O’Dell Tech Innovation Hub, “About O’Dell Tech Innovation Hub,” <https://innovationhub.odelltech.com/about/>, Accessed: Sep. 15, 2026.
- [2] O’Dell Tech, “Odelltech,” <https://odelltech.com/>, Accessed: Sep. 15, 2026.
- [3] O’Dell Tech Innovation Hub, “Home,” <https://innovationhub.odelltech.com/>, Accessed: Sep. 15, 2026.
- [4] Python Software Foundation, “The Python Tutorial,” *Python 3 documentation*, <https://docs.python.org/3/tutorial/>, Accessed: Sep. 15, 2026.
- [5] Django Software Foundation, “Models,” *Django 4.0 documentation*, <https://docs.djangoproject.com/en/4.0/topics/db/models/>, Accessed: Sep. 15, 2026.
- [6] Django Software Foundation, “Using the Django authentication system,” *Django 4.0 documentation*, <https://docs.djangoproject.com/en/4.0/topics/auth/default/>, Accessed: Sep. 15, 2026.
- [7] S. Chacon and B. Straub, “Getting started—about version control,” in *Pro Git*, 2nd ed., sec. 1.1, <https://git-scm.com/book/en/v2/Getting-Started-About-Version-Control>, Accessed: Sep. 15, 2026.
- [8] Oracle, “MySQL 8.4 Reference Manual: FOREIGN KEY Constraints,” sec. 15.1.20.5, <https://dev.mysql.com/doc/refman/8.4/en/create-table-foreign-keys.html>, Accessed: Sep. 15, 2026.

- [9] OWASP Foundation, “Authorization Cheat Sheet,” *OWASP Cheat Sheet Series*, https://cheatsheetseries.owasp.org/cheatsheets/Authorization_Cheat_Sheet.html, Accessed: Sep. 15, 2026.
- [10] Django Software Foundation, “Database transactions,” *Django 4.0 documentation*, <https://docs.djangoproject.com/en/4.0/topics/db/transactions/>, Accessed: Sep. 15, 2026.
- [11] Django Software Foundation, “QuerySet API reference (select for update()),” *Django 4.0 documentation*, <https://docs.djangoproject.com/en/4.0/ref/models/querysets/>, Accessed: Sep. 15, 2026.
- [12] Django Software Foundation, “Writing and running tests,” *Django 4.0 documentation*, <https://docs.djangoproject.com/en/4.0/topics/testing/overview/>, Accessed: Sep. 15, 2026.
- [13] International Software Testing Qualifications Board, “Certified Tester Foundation Level Syllabus,” ver. 4.0.1, Sep. 2024, https://istqb.org/?download_id=3345&sdm_process_download=1, Accessed: Sep. 15, 2026.
- [14] Green University of Bangladesh, Department of Computer Science and Engineering, “Syllabus of B.Sc. in Computer Science and Engineering, 2019,” Program Outcomes, p. 4, Approved Dec. 19, 2019, https://siteadmin.green.edu.bd/uploads/documents/2_1709286695_documents.pdf, Accessed: Sep. 15, 2026.
- [15] United Nations, “The 17 Goals,” Department of Economic and Social Affairs, Sustainable Development, <https://sdgs.un.org/goals>, Accessed: Sep. 15, 2026.